

ML Challenge 2026: Business Entity Resolution Solution Template

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1. Executive Summary

- **Pipeline.** We match each Source-1 business to its Source-2/3 copies with a precision-first pipeline:
 - per-country multi-view blocking;
 - three gradient-boosted scoring stages on name, address and legal-form evidence;
 - fine-tuned multilingual cross-encoders on the uncertain pairs: e5, bge, and LoRA-tuned Qwen2.5-1.5B/7B;
 - a per-S1 decision that maximises expected F0.5.
- **France.** It appears only in the test set.
 - We self-trained on our own guarded, cross-fitted French decisions, for two rounds.
 - An independent Qwen2.5-7B re-check of the confident predictions removed French look-alike decoys that string models accept.
- **Scores.** 0.9913 macro F0.5 on a labelled US/India holdout, and 0.990879 on the public leaderboard.

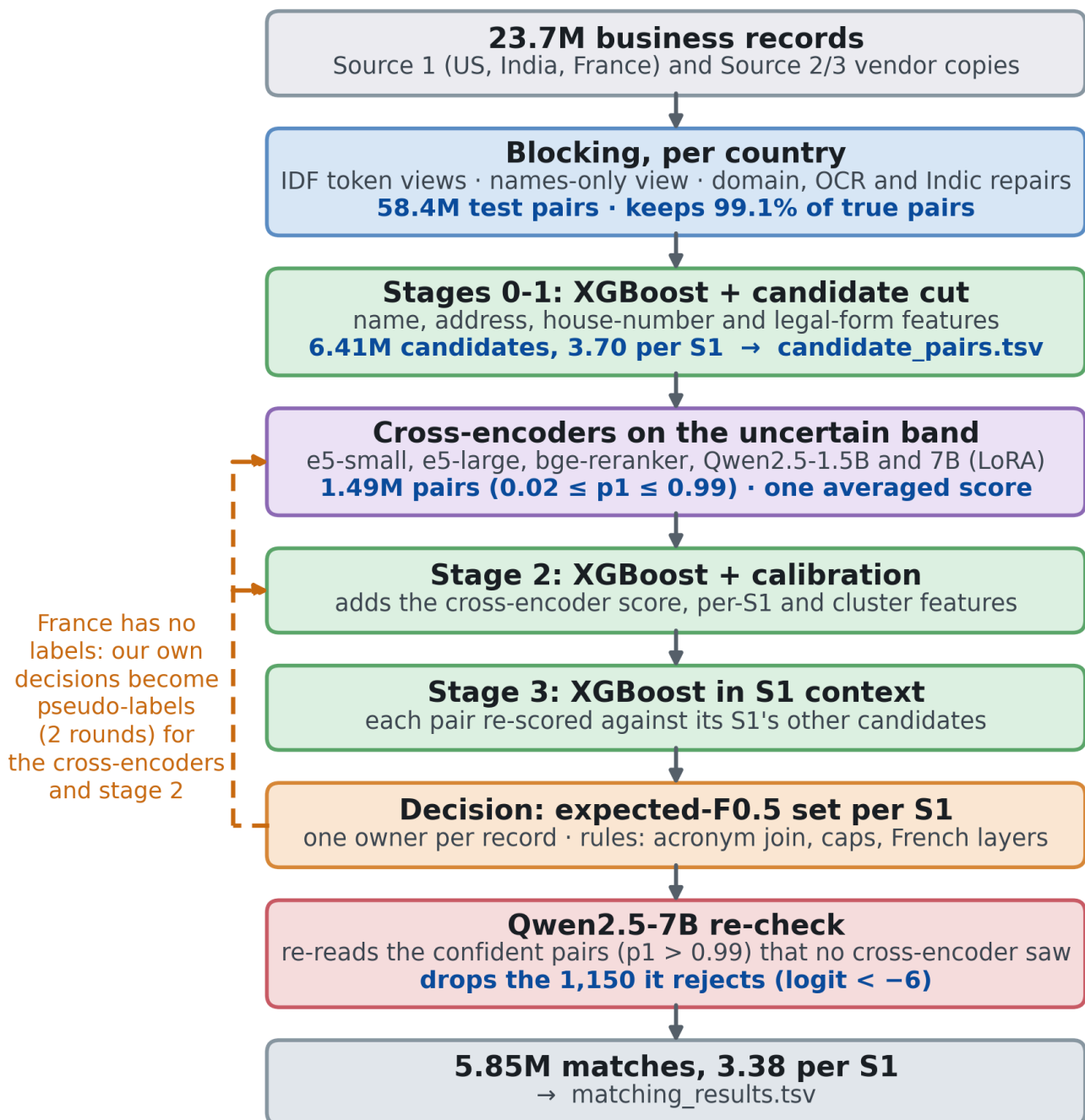


Figure 1. The pipeline, with test-set counts at each stage. The dashed loop is the French self-training (§2.2).

Rules we kept:

- **Only the provided data.** No external databases, APIs or look-ups.
- **Hand-written lexicons** are spelling equivalences only:
 - legal forms, street types, state and region names;
 - list-append words, the OCR digit map, ordinal words.
- **Models:** every one is MIT or Apache-2.0 and at most 8B parameters. The largest is Qwen2.5-7B, with 7.6B.

2. Methodology

2.1 Problem Analysis

The task, as the metric sees it:

- Each S1 business has 0–11 noisy vendor copies in S2/S3, 3.46 on average.
- About 5.6% have none. For those, an empty prediction scores 1 and any prediction scores 0.
- F0.5 weights precision twice as much as recall. So a wrong pair costs more than a missed one, and we designed for precision first.

How true copies differ from the original (measured on labelled pairs):

- case and accent changes;
- legal forms reformatted or dropped: LLC / Inc / Pvt Ltd, and in France SARL / SAS / EURL in about 23% of copies;
- word reorders and list appends ("& Fils", "Services");
- acronyms, typos and whole-word garbles;
- street-type abbreviations (Rue → R., Avenue → Av);
- city-first ordering, and region or department swaps;
- house-number changes, and empty addresses.

The blocking keys (§3) and the features (§4) are built around these edits.

Decoys:

- The test set has about 23% more records per S1 than train, while true matches per S1 stay flat. So the extra records are look-alike decoys:
 - a business word swapped at the same address;
 - a nudged house number;
 - the same name at another address.
- 46–54% of S1 share their name with another S1, so the address has to decide.
- Our response:
 - features for each look-alike edit;
 - candidates compete per S1 in a set-level decision;
 - every record is owned by at most one S1.

France (test only, no labels):

- **Generic names** (city + category word, e.g. "lille club sarl"). Each French decoy we removed shares its name with a median of 43 other French S1.
- **Shared addresses:** about 11% of French S1 share their exact address with another S1.
- **Administrative names:** about 32% of French addresses carry department or region names.
- **What our first uploads showed.** They scored 0.976–0.980 on the leaderboard, against 0.984–0.989 on the US/India holdout, which implied France at about 0.93. That gap is why most of our later work targets France:
 - per-country statistics and French address normalisation;
 - self-training;
 - a second opinion on confident pairs.

2.2 Solution Strategy

Approach Type: Hybrid.

- Multi-key blocking → gradient-boosted pair classifiers (three stages) → transformer cross-encoders on the uncertain band → per-S1 expected-F0.5 set selection → rules → an LLM re-check of confident predictions.

Core Innovation:

1. Self-training for a country without labels.

- French pairs are pseudo-labelled from our best model's final decisions: positive if kept with calibrated $p \geq 0.9$, negative if rejected with $p \leq 0.05$, otherwise unlabelled.
- The labels are guarded: rule-derived populations override them, and empty-address records keep round-1 labels.
- They are cross-fitted by S1 group, so no French pair is scored by a model that saw its own label.
- Two rounds, each taught by the previous best model.

2. A set-level decision. For each S1 we choose the prediction set with the highest *expected* F0.5 under calibrated probabilities, instead of thresholding each pair.

3. An independent re-check of confident predictions.

- 94.5% of final predictions have stage-1 $p_1 > 0.99$, so no cross-encoder ever read them.
- A LoRA-tuned Qwen2.5-7B re-reads them and removes the pairs it rejects ($\text{logit} < -6$).
- The cut-off was fixed on labelled data first.

How we decided what to keep:

• US/India.

- A fixed shared holdout: 25% of train S1 (549,699 S1), never used for fitting, scored with the official macro F0.5.
- A component was added only when a paired bootstrap on this holdout showed a gain. Ties went to the simpler option.
- Post-hoc rules also had to gain on both fixed halves of the holdout.

• France.

- Label-free checks, e.g. predicted matches per S1 against the generator's limit.
- Estimates corrected with the 7B's judgement.
- Leaderboard probes that changed only France.

3. Candidate Generation (Blocking)

Blocking keys used (blocking v3, `ber.block`):

- **Per-country search.** Every search runs per exact country label, so IDF weights and neighbours are never borrowed across countries. Records with an empty label join every partition.
- **Token view:**
 - IDF-weighted token overlap over name + address, in both directions: each S1's top 40 records and each record's top 8 S1;
 - a pair is kept if the record is in the S1's top 15 or the S1 is in the record's top 4;

- name-word compound keys, and a (house number, street word) key that skips street types, so French addresses match.
- **Names-only view**, for records whose address is empty or has at most 3 tokens:
 - character 4-grams, which catch garbles;
 - single-token names of 8+ letters, which catch domains and handles such as `renterianaborweddle.com`.
- **Repairs before matching:**
 - domain and handle names segmented into the country's S1 words;
 - OCR digits in names repaired;
 - an Indic→Latin token dictionary (693 entries, e.g. `kanstrakshan` → `construction`), learned only from true pairs of the training folds.

Candidate pairs generated:

- **Retrieval:** 66.4M train and 58.4M test pairs.
- **Cut:**
 - a stage-0/1 gradient-boosted scorer keeps pairs with $p1 \geq 0.02$;
 - for each record it also keeps its two highest-scoring S1;
 - acronym joins are added.
- **Submitted file:** `candidate_pairs.tsv` has **6,410,308 pairs, 3.70 per test S1**.

How you ensured true matches were not lost:

- **Measured, not assumed.** We tracked blocking recall on the holdout. **99.1% of true holdout pairs are candidates:** 16.5k of about 1.9M are missed, mostly empty-address records whose name was changed.
- **One key per noise type:**
 - n-grams catch garbles;
 - the domain key catches web-style names;
 - house numbers and address words catch renamed businesses;
 - transliteration catches Indic script.
- **The cut was tested.** Tighter candidate cuts (top-1 per record; $p1 \geq 0.05$) lost -0.000106 and -0.000041 holdout F0.5, so we kept the more generous cut.

4. Matching Model

Features used. All IDF statistics are fitted per country; there is no country feature.

group	features
Name (normalised core names, legal forms stripped)	character ratio, token-sort, token-set, partial ratio, Jaro-Winkler · token Jaccard and containment both ways · IDF-weighted token cosine · first/last-token agreement · space-free similarity for domains and hashtags · Indic-script mismatch · legal form equal / compatible / conflict / missing, plus an 8-feature legal-form group · "unexplained token" counts and IDF mass on each side, with an edit-distance substitution flag · learned look-alike odds (per-token log-odds from training folds only)

group	features
Address and house number	house number: equal, truncation, suffix, letter/bis suffix, nudge (1–10), absolute and relative difference, digit edit distance, set Jaccard, S1-only / record-only numbers, unit equality · edit profile: ± 1 move, digit substitution, digit swap, suffix, length difference · street-token Jaccard and IDF overlap, street Jaro-Winkler · city and state equality · whole-address token-set and character 3-gram cosine · empty / short / landmark / PO-box / null flags, token counts
Context	retrieval: the pair's score and rank in each blocking view, and how many views found it · rivalry: gap to the S1's best candidate, margin over any other S1 for the same record · log-counts of candidates and of S1 sharing the name (counts, not rates, so France's smaller pool is not over-weighted) · per-S1 aggregates of p1: expected cluster size, counts above thresholds, rank · cluster support: similarity to the S1's other confident records · source (S2/S3) · the cross-encoder score (below)

Model type:

- **XGBoost (GPU), four stages.** Each stage is out-of-fold over three S1 groups, so the next stage learns from honest scores. Seeds are fixed.
 - Stage 0 prunes about 80% of pairs cheaply.
 - Stage 1 gives p1, which sets the candidate cut and the cross-encoder band.
 - Stage 2 adds the cross-encoder, cluster and per-S1 features. It also adds the French pseudo-labelled rows: cross-fitted, with weight 3 in the model behind France's predictions and weight 1 in the US/India model.
 - Stage 2 is isotonic-calibrated on out-of-fold scores.
 - Stage 3 re-scores each pair in the context of its S1's other candidates.
- **Cross-encoders** run only on the uncertain band, $0.02 \leq p1 \leq 0.99$: 1.57M train and 1.49M test pairs. That's where reading the text pays.
 - Input: `name` ; `address` of both sides, at most 96 tokens.
 - Each is three out-of-fold models.
 - The French-facing members also train on French pseudo-labelled pairs, cross-fitted by S1 third.

model	size	licence	training	band AUC (holdout)
stage-1 p1, for reference				0.930
multilingual-e5-small	118M	MIT	full fine-tune, 1 epoch, lr 5e-5	
multilingual-e5-base (round-1 French teacher only)	278M	MIT	full fine-tune	
multilingual-e5-large (+ French self-trained)	560M	MIT	full fine-tune, 1 epoch, lr 2e-5, batch 128	0.939 (0.944)
bge-reranker-v2-m3 (+ French self-trained)	568M	Apache-2.0	same as e5-large	0.942 (0.942)
Qwen2.5-1.5B, French self-trained	1.5B	Apache-2.0	LoRA	0.938

model	size	licence	training	band AUC (holdout)
Qwen2.5-7B, French self-trained	7.6B	Apache-2.0	LoRA	0.944

- **The Qwen models** use a one-logit classification head with LoRA:
 - $r=16$, $\alpha=32$, dropout 0.05, on all attention and MLP projections;
 - bf16, AdamW, lr $1e-4$ with 3% warm-up, batch 64, 1 epoch.
 - The 7B trained on three H100s, one out-of-fold model per GPU, in about 2 hours.
- **Why a mix, not the biggest model:** the 7B's band AUC (0.944) only matches the self-trained e5-large (0.944). Its value is diversity: a different model family in the mix, and an independent reader of confident pairs.
- **The mix.** The z-scored logits are averaged into one stage-2 feature. Separate logits made stage 2 extrapolate where the models disagree, which is $4\times$ as common in France.
 - US/India: e5-large, French-self-trained e5-large, bge, Qwen2.5-1.5B, and Qwen2.5-7B counted twice.
 - France: Qwen2.5-1.5B, two e5-large seeds and bge, all self-trained, plus a France-weighted e5-large.

Threshold selection method:

- **Expected-F0.5 set selection.** Calibrated stage-3 probabilities feed a dynamic programme over each S1's owned candidates. It picks the prefix with the highest expected F0.5.
 - It replaced a tuned global threshold after a paired bootstrap on the holdout: $+0.000048$ [$+0.000007$, $+0.000091$], together with the acronym join and caps.
 - Settings tuned on the holdout:
 - logit shift $+0.2$;
 - -0.3 for records contested by 4 or more S1;
 - a 0.01 expected true match outside the candidates, a recall term.
- **Ownership:** each record goes to its highest-scoring S1.
- **Rule layers:**
 - acronym join;
 - per-source caps;
 - for France:
 - an exact-address copy rule, 99.99% precise on US/India;
 - a cross-commune drop;
 - a look-alike word-swap drop;
 - the same expected-F0.5 set selection applied to France's stage-3 probabilities, where France had used a threshold until then.
- **The 7B re-check cut-off (−6).** Chosen on the holdout, where it gave the largest gain: $+0.000037$, positive in both fixed halves and in 99.7% of 3,000 random holdout subsets. Looser cut-offs gained less (−5: $+0.000029$; −4: $+0.000017$), and from −3 on they lose.

5. Results & Error Analysis

F_{0.5} Score (macro):

evaluation	macro F0.5
labelled holdout (US/India, 549,699 S1)	0.9913 (US 0.9911, India 0.9916); micro precision 99.9%, recall 97.5%
public leaderboard (US, India and France)	0.990879

Where the score came from. Each step is a leaderboard submission built on top of the previous one.

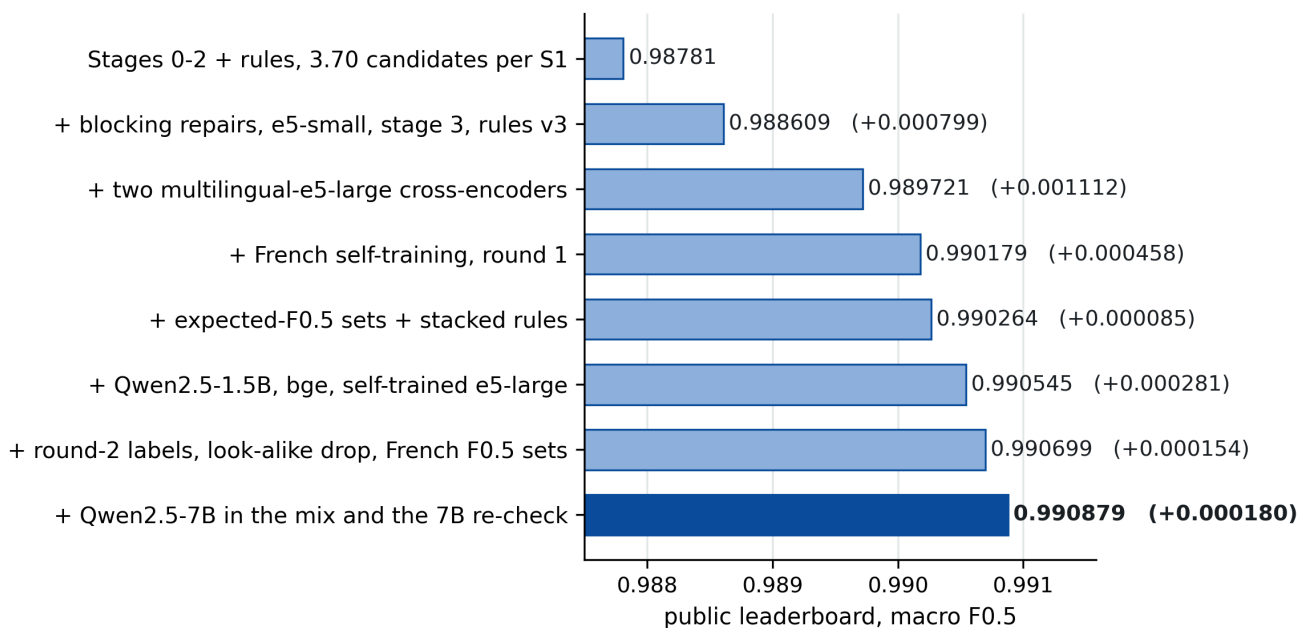


Figure 2. Public leaderboard macro F0.5 after each step, with the gain over the previous step. The last bar is the submission in this package.

Common false positives (wrong merges):

- **Generic-name decoys (France)**, e.g. lille ecole sarl | 42 rue gutenber | 42 q. du wault: same name and house number, different street.
 - On labelled data this pattern is 0.5% true where our model rejects it, and 99.7% true where it accepts it.
 - String features cannot separate the two cases; the 7B could.
 - Removing its 840 French rejects gave an estimated +0.00011 of our last +0.00018.
- **Business-word swaps at the same address** ("antenne danse" vs "antenne gaz").
- **Near-identical short names** ("osd" vs "otd").
- **Chain branches** sharing one name.

Common false negatives (missed matches):

- **Where the loss is.** On the US/India holdout the model is 99.9% precise, so the remaining loss is mostly recall:
 - 69% comes from partially-missed S1;
 - 20% from S1 missed entirely;

- 8% from false positives.
- **Which pairs are missed:** mostly empty-address copies with changed or invented names. They either never become candidates (16.5k) or are scored near 0 (15.2k).
- **Garbles:** heavy garbles also cause misses, e.g. the French "EHPAD" typed as "ehpvd" or "phpad".
- **We did not add recall rules.** Every one we tried was 17–71% precise, below the ~75% a match needs to help F0.5:
 - empty-address name match;
 - an alias at the same address;
 - the top candidate of an empty S1;
 - 7B-confirmed additions.

6. Conclusion

- **Results.** A precision-first design with calibrated, set-level decisions brought the labelled countries to 0.9913. Guarded self-training carried France, which has no labels, most of the way.
- **The 7B re-check.** An independent re-check of the "obvious" predictions found decoys that feature models accept.
- **What we learned:**
 - Self-training paid for two rounds. A third round did not beat them without the same French decision layers.
 - Without labels, a probability-based estimate needs a second, independent judge, because it inherits the model's blind spots.

Appendix

A. Code Artefacts

`code/business_entity_resolution/` regenerates both output files from the raw train/test TSVs.

path	contents
<code>reproduce.sh</code>	the entry point. <code>VARIANT=compositeB</code> (the default) runs <code>src/box/compositeB.sh</code>
<code>src/ber/</code>	the team package: I/O, IDs, normalisation, blocking, context features, output writer, evaluation (holdout split, macro F0.5, paired bootstrap). One CLI: <code>python -m ber.pipeline --stage <stage> --split <split> --tag <tag></code>
<code>src/model_v1/</code>	pair features, stages 0–3, cross-encoder training (e5/bge; Qwen LoRA), self-training labels, the decision, the rules, the stacked rules and France layers, <code>pipeline/france_mixmdp.sh</code> (France)
<code>src/box/</code>	the Composite B driver, the Qwen2.5-7B out-of-fold training (<code>llm_group.py</code> , <code>llm_merge.py</code>), the 7B re-check (<code>rescore_export.py</code> , <code>score_pairs.py</code>), and the per-country composition (<code>compose_tsv.py</code>)
<code>tests/</code> , <code>requirements.txt</code> , <code>README.md</code>	unit tests, pinned versions, exact run instructions

How to run (from the unzipped package; details in `README.md`):

```
pip install -r code/business_entity_resolution/requirements.txt && pip install -e
code/business_entity_resolution
MODEL_DIR="$PWD/code/business_entity_resolution/src/model_v1" BER_OUTPUT_DIR="$PWD/output_rerun" \
  VARIANT=compositeB bash code/business_entity_resolution/reproduce.sh
```

What the driver runs:

1. The US/India chain: records → blocking → features → stages 0–1 → cross-encoders → stages 2–3 → decision → rules. The round-1 French teacher is built along the way.
2. French pseudo-labels from that model.
3. Qwen2.5-7B training (three out-of-fold groups).
4. The US/India model with the 7B in the mix.
5. The 7B re-check of confident predictions.
6. The France block (round-2 guarded labels ×3 and the French layers).
7. Per-country composition, the 7B drops, the validator and a strict audit.

Hardware and time:

- **CPU stages:** 24+ cores, 32 GB RAM.
- **Cross-encoders:** one H100 80 GB.
- **Qwen2.5-7B:** three 80 GB GPUs, about 2 hours.
- **Per-step times** are in `README.md`; the GPU stages dominate.
- **Software:** Python 3.12–3.13, torch 2.11, transformers 5.17, peft 0.21, xgboost 3.2.0.
- **Model revision:** Qwen2.5-7B `d149729398750b98c0af14eb82c78cfe92750796`.

Reproducibility:

- Seeds are fixed, and every artifact records its command and git commit.
- GPU training is not bit-deterministic across machines.
 - Rebuilding an earlier model on different hardware gave 0.991261 holdout F0.5, against 0.991246 originally.
 - It moved 1–4 predictions per 1,000 S1.
- So a rerun lands within about ± 0.0001 . `output/` holds the exact submitted bytes, and the final composition reproduces `matching_results.tsv` byte-for-byte from the saved intermediates.

B. Additional Results

Ablations (labelled US/India holdout; paired bootstrap):

change	effect on macro F0.5
expected-F0.5 set selection vs a tuned threshold	+0.000048 [+0.000007, +0.000091]
Qwen2.5-7B ×2 in the cross-encoder mix	India +0.000066 (P 0.998), US +0.000026
7B re-check (drop logit < -6)	+0.000037, both halves positive
tighter candidate cut (top-1 / p1 ≥ 0.05)	-0.000106 / -0.000041
French self-training, round 1 / round 2 (leaderboard)	+0.00046 / +0.00015

The cross-encoder mix (holdout after stage 3 and the decision; only the mix changes):

mix	overall	US	India
e5-large, e5-large self-trained, bge, Qwen2.5-1.5B	0.991261	0.991064	0.991558
+ Qwen2.5-7B ×1	0.991276	0.991076	0.991577
+ Qwen2.5-7B ×2 (used)	0.991323	0.991114	0.991635
+ Qwen2.5-7B ×3	0.991322	0.991111	0.991639
Qwen2.5-7B alone	0.991286	0.991089	0.991581

The 7B re-check. The table covers predictions with $p_1 > 0.99$ that no cross-encoder had read: 631,001 predictions on a labelled 34% holdout sample, plus the French test predictions.

7B logit	holdout predictions	truly a match	French predictions
< -6	24	8.3%	859
[-6, -4)	51	86.3%	209
[-4, -2)	138	94.9%	321
[-2, 0)	6,616	99.6%	897
[0, 2)	16,893	99.8%	4,461
≥ 2	579,440	100.0%	780,629

- **Rejection rates:** the 7B rejects 0.10% of French predictions, against 0.004–0.007% of US/India ones.
- **Leakage check:** the rejection rate is the same in each third of French S1 (0.107–0.112%), including the third the model never saw labels for.

Controls on the leaderboard:

- **Composite B with the round-3 French model, without the French decision layers:** 0.990833, -0.000046.
 - By our 7B-corrected estimate, the missing French expected-F0.5 decision accounts for about +0.00005.
 - The round-3 model itself was slightly better.
- **Composite B with the French 7B drops extended below -6** (1,699 more drops, 251 look-alikes restored): 0.990875, a tie. The -6 cut-off was already right.

What did not work (measured, then dropped):

- **Recall rules**, 17–71% precise:
 - empty-address name matches;
 - aliases at the same address;
 - rescuing an empty S1's top candidate;
 - 7B-confirmed additions.
- **A learned blend of all six cross-encoder logits** as the decision score: band AUC 0.9575 against 0.9276, but -0.001 macro F0.5.
- **A decision tree mining extra drops:** +0.000033 on one half, -0.000015 on the held-out half.

- **Copy-count tie-breaking** for empty-address records: owner accuracy 0.31, against 0.27 by chance. An add rule on it loses about 0.00057.
- **Renormalising French probabilities per record:** -0.000007 to -0.000137 .
- **Other encoders:**
 - Qwen3-4B scored below the 7B (band AUC 0.941).
 - mDeBERTa-v3 hit non-finite bf16 gradients.
 - gte-multilingual-reranker's remote code crashed under our transformers version.